

3.2. Patchscopes Encompasses Prior Methods

Many recent methods inspect LLM representations by projecting them to the output vocabulary space (nostalgebraist, 2020; Din et al., 2023; Belrose et al., 2023). Formally, an estimation of the output distribution is obtained from the representation $\mathbf{h}_i^\ell$ at position i and layer ℓ by:

$$\mathbf{p}_i^\ell = \text{softmax}(W_U f(\mathbf{h}_i^\ell)) \in \mathbb{R}^{|V|},$$

where $W_U \in \mathbb{R}^{|V| \times d}$ is the model’s unembedding matrix and f is a simple mapping function, such as the identity function or an affine mapping. We note that the operation applied to $f(\mathbf{h}_i^\ell)$ is the same computation applied by the model to the last-layer representation for obtaining the next-token prediction. Therefore, prior methods that inspect representations in the vocabulary space can be viewed as a class of Patchscopes that maps representations from any source layer ℓ to the last target layer L^* . Differences between these methods lie in the choice of f ; logit lens (nostalgebraist, 2020; Dar et al., 2023) applies the identity function, linear shortcuts (Din et al., 2023) uses a linear mapping function, and tuned lens (Belrose et al., 2023) trains an affine mapping. Recently, Hernandez et al. (2023b) introduced LRE Attribute Lens that builds f based on a relation linearity assumption, and showed its effectiveness in attribute extraction.

This class of methods has proven to be effective for different applications, for example, in improving inference efficiency via early exiting (Din et al., 2023). While the majority of methods and applications in this category use a single model ($\mathcal{M}^* = \mathcal{M}$), Merullo et al. (2022) had demonstrated successful caption generation with a generative image model as $\mathcal{M}$ and a language model as $\mathcal{M}^*$.

Another category of inspection methods intervene on the LLM computation. Contemporary to our work, Pal et al. (2023) have investigated whether it is possible to anticipate multiple generated tokens ahead from a given hidden representation, rather than estimating just the next-token prediction. Their method (Future Lens) uses a target prompt different from the original prompt (i.e., $T \neq S$) and is designed to decode subsequent tokens from information encoded in a hidden representation $\mathbf{h}_i^\ell$. Example target prompts are “The multi-tokens present here are ” and “Hello! Could you please tell me more about ”. Future Lens can be cast as another Patchscope with $\mathcal{M}^* = \mathcal{M}$ and $\ell^* = \ell$.

More broadly, Patchscopes also cover recent mechanistic interpretability methods that analyze internal processes in LLMs with inference computation interventions. Specifically, causal tracing (Meng et al., 2022a) uses a source prompt augmented with Gaussian noise as the target prompt. Other previous methods have intervened on one or more target layers during inference by patching zero vectors to the computation (Wang et al., 2022; Conmy et al., 2023; Geva

et al., 2023), namely, setting $f(\mathbf{h}) = \mathbf{0}$. For a configuration summary of how these interpretability methods can be cast as Patchscope instances, see §A, Tab. 4.

3.3. Patchscopes Enables Novel Inspection Methods

Prior work has utilized specific patching configurations for interpretability, largely patching the same model while using the same prompt (i.e., $\mathcal{M}^* = \mathcal{M}$, $T = S$). The framing of Patchscopes introduces a wide range of unexplored setups potentially unlocking new inspection capabilities.

Specifically, we observe that modifying the target prompt enables an expressive decoding of a wide range of features, detached from the source prompt computation. For instance, we can use the prompt “The capital of X is” to check if the capital city of a given country is extractable from its (last token) hidden representation at a specific layer. Similarly, a prompt like “Tell me facts about X ” can be leveraged to assess whether the model has resolved the entity name in a specific layer. Contrary to probing, this approach is not restricted by the number of classes of the chosen feature.

Moreover, when the inspected model is not expressive enough to answer certain queries, patching representations into a more capable model could be useful (Hernandez et al., 2022; Singh et al., 2023; Schwettmann et al., 2023).

4. Experiments

In this section, we evaluate our framework on decoding next-token predictions (§4.1), extracting attributes (§4.2), analyzing the contextualization of entity names (§4.3), and leveraging stronger models for inspection via cross-model patching (§4.4). See a summary in Tab. 1.

4.1. Decoding of Next-Token Predictions

As introduced in §3.2, let $\mathbf{p}^L$ be the output probability distribution for some input, obtained by multiplying the final-layer last-position hidden representation $\mathbf{h}^L$ by the unembedding matrix $W_U \in \mathbb{R}^{|V| \times d}$. We wish to estimate $\mathbf{p}^L$ from intermediate representations $\mathbf{h}^\ell$ s.t. $\ell < L$. Particularly, we ask how early in the computation the model has concluded its final prediction from the given context. In our experiments, we consider multiple LLMs – LLaMA2 (13B) (Touvron et al., 2023b), Vicuna (13B) (Chiang et al., 2023), GPT-J (6B) (Wang & Komatsuzaki, 2021), and Pythia (12B) (Biderman et al., 2023) (see more details in §B.1).

Methods We compare vocabulary projection methods (§3.2) with a new Patchscope. Each method yields an estimated output probability $\tilde{\mathbf{p}}^\ell$ by patching an intermediate representation $\mathbf{h}^\ell$ to the model’s final layer. Here, we focus on the common setting where $\mathcal{M} = \mathcal{M}^*$, and discuss extensions to $\mathcal{M} \neq \mathcal{M}^*$ in §4.4.

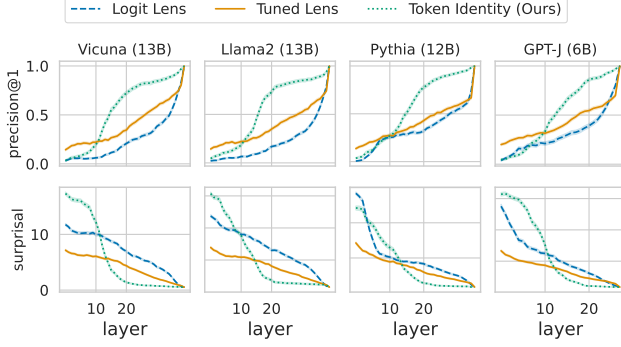


Figure 2. Precision@1 (↑ is better) and Surprisal (↓ is better) of next-token prediction estimation in multiple models. From layer 10 and upwards, the token identity method (ours) consistently outperforms the rest of the baselines across all the models.

- **Logit Lens:** Following prior work (nostalgebraist, 2020; Geva et al., 2022a), we define f as the identity function, meaning no change is applied to the patched representation. That is, $f(\mathbf{h}) := \mathbb{I}(\mathbf{h})$.
- **Tuned Lens:** Motivated by Belrose et al. (2023); Din et al. (2023), we employ an affine mapping function between representations at layer ℓ and the final layer L . Specifically, we feed the model examples from a training set $\mathcal{T}$ and for each example $s \in \mathcal{T}$ obtain a pair $(\mathbf{h}_s^\ell, \mathbf{h}_s^L)$ of hidden representations. Then, we fit a linear regression model to find the matrix $A^\ell \in \mathbb{R}^{d \times d}$ and the bias vector $\mathbf{b}^\ell \in \mathbb{R}^d$ that are numerical minimizers for $\sum_{s \in \mathcal{T}} \|\mathbf{A}^\ell \mathbf{h}_s^\ell - \mathbf{h}_s^L + \mathbf{b}^\ell\|^2$. We define f as: $f(\mathbf{h}^\ell) := \mathbf{A}^\ell \mathbf{h}^\ell + \mathbf{b}^\ell$.
- **Token Identity Patchscope:** Unlike the previous methods, here we use a target prompt that is different from the source prompt ($T \neq S$) and is meant to encourage decoding the token identity of the hidden representation. Also, while the above methods skip the computation between layers l and L , here we modify it such that all the information from the source prompt computation is discarded, except for the patched representation. We craft a prompt with k demonstrations representing an identity-like function, formatted as “tok₁ → tok₁; tok₂ → tok₂; ...; tok_k”. See §B.3 for further details and an experiment showing this method’s robustness to different demonstrations. Note that this Patchscope does not require any training.

Evaluation Following Din et al. (2023), we use Pile evaluation set and the following metrics (see details in §B.2):

- **Precision@1** (↑ is better): The portion of examples for which the highest-probability token t in the estimated probability distribution matches the highest-probability token in the original output distribution. That is, if

$$\arg \max_t (\tilde{\mathbf{p}}_t^\ell) = \arg \max_t (\mathbf{p}_t^L).$$

- **Surprisal** (↓ is better): The minus log-probability of the highest-probability token in the predicted distribution $\tilde{\mathbf{p}}^\ell$ according to $\mathbf{p}^L$, i.e., $-\log \mathbf{p}_{\tilde{t}}^L$, where $\tilde{t} = \arg \max_t (\tilde{\mathbf{p}}_t^\ell)$.

Results Across all the models, from layer 10 and upwards, the token identity Patchscope consistently outperforms the other baselines, obtaining a gain of up to 98% in layers 18-22 (see Fig. 2). This demonstrates the utility of leveraging the model’s decoding procedure for inspecting representations of different source prompts, and shows that *in most cases* hidden representations in early layers carry the prediction information regardless of their context.

In the first 10 layers, performance of all methods is worsened, with the token identity prompt performing on-par with logit lens, and tuned lens performing slightly better, which could be due to the additional training of its mappings. Low performance in these layers is expected, as it is where the input contextualization happens. In §4.3, we introduce a Patchscope geared towards unraveling this process.

4.2. Extraction of Specific Attributes

Classification probes are arguably the most commonly used method for checking if certain attributes are encoded in hidden representations (Belinkov, 2022; Belinkov & Glass, 2019). However, they need to be trained, and the range of attribute classes needs to be known a priori. Here we show that repurposing Patchscopes for attribute extraction overcomes these limitations. First, it does not require training. Second, it is not limited by a predefined set of labels, but rather benefits from an open vocabulary. In addition, by taking advantage of the model’s nonlinearities, it can capture more complex relations compared to linear probes.

Experimental Setup Consider factual and commonsense knowledge represented as triplets (σ, ρ, ω) of a subject (e.g., “United States”), a relation (e.g., “largest city of”), and an object (e.g., “New York City”). We investigate to what extent the object ω can be extracted from the last token representation of the subject σ in an arbitrary input context. To this end, we conduct experiments on 8 commonsense and 25 factual knowledge tasks curated by Hernandez et al. (2023b). This dataset includes (σ, ρ, ω) triplets for different relations, along with prompt templates that verbalize them in natural language. We conduct experiments with GPT-J (6B) (Wang & Komatsuzaki, 2021), filtering the data to keep only the examples where ω appears in the the model’s continuation of the prompt up to 20 tokens. The choice of 20 balances computation cost with accommodating the open-ended nature of Patchscopes, as the ground truth token does not necessarily appear in the next immediate token in a fluent

Table 2. Feature extraction accuracy (mean $\pm$ std). Comparing zero-shot feature extraction Patchscope to a logistic regression probe shows that despite using *no training data*, it has a significantly higher accuracy than baseline in 6 out of 12 tasks. We use pairwise t-test with Bonferroni correction for comparing the two methods. ** and * indicate $p < 1e-5$ and $p < 1e-4$, respectively.

	Task	Probe	Patchscope
Commonsense	Fruit inside color	37.4 ± 6.6	38.0 ± 18.7
	Fruit outside color	35.5 ± 3.1	$71.0 \pm 13.3^{**}$
	Object superclass	$65.6 \pm 10.5^*$	54.8 ± 11.3
	Substance phase	73.8 ± 3.7	$91.9 \pm 1.7^{**}$
	Task done by tool	10.1 ± 3.2	$48.1 \pm 13.2^{**}$
Factual	Company CEO	5.0 ± 2.6	$47.8 \pm 13.9^{**}$
	Country currency	17.7 ± 2.2	$51.0 \pm 8.9^{**}$
	Food from country	5.1 ± 3.7	$63.8 \pm 11.3^{**}$
	Plays pos. in sport	75.9 ± 9.1	72.2 ± 7.2
	Plays pro sport	53.8 ± 10.3	46.3 ± 14.2
	Product by co.	58.9 ± 7.2	63.2 ± 10.7
	Star constellation	17.5 ± 5.3	18.4 ± 5.1

response. For each example, we sample 5 utterances from the WikiText-103 dataset (Merity et al., 2016) that include σ and use them as S . Lastly, we keep tasks with at least 15 samples, which results in 5 commonsense and 7 factual tasks with a total of 1,453 datapoints. See details in §C.

Methods We compare our proposed Patchscope against linear probing (Köhn, 2015; Gupta et al., 2015).

- **Zero-shot Feature Extraction** Patchscope: We craft T as a general verbalization of ρ followed by a placeholder for σ , such that $i^* = m$. For example, we use $T \leftarrow$ “The largest city in x” with “x” as a placeholder for the subject. To extract the object from the entity representation in S , we patch the representation of token “x” at layer ℓ^* with the representation of “States” from layer ℓ , and consider if the generated text includes ω . The remaining configurations of this Patchscope are $f \leftarrow \mathbb{I}, \mathcal{M}^* \leftarrow \mathcal{M}, i \leftarrow$ the last token of σ in S . We consider all combinations of $\ell \in [1, \dots, L] \times \ell^* \in [1, \dots, L^*]$. Later in this section, we discuss the role of ℓ pertaining to attribute extraction.
- **Logistic Regression Probe:** Let Ω represent the range of possible objects for a given relation. We use the set of unique values of ω in the training set as a proxy for Ω . We train a logistic regression probe (Köhn, 2015; Gupta et al., 2015) for each layer that predicts $\omega \in \Omega$ from the last token representation of σ . Given that 6 out of 12 tasks have fewer than 40 datapoints, we use three-fold cross-validation for training and evaluation of this baseline. Note that we have excluded tasks where

the probe fails completely due to insufficient number of training examples (fewer than 15 datapoints).

Evaluation We measure the average attribute extraction accuracy. For a given sample, the Patchscope is considered correct if $\exists \ell^* \in [1, \dots, L^*]$ where the generated text up to 20 tokens includes ω . For the probe, a prediction is correct if the highest probability is assigned to ω .

Results Tab. 2 summarizes the results, averaged over $\ell \in [1, \dots, L]$. We conduct a T-test with Bonferroni correction to compare the two methods. Despite using no training data and having no restrictions on the output, the Patchscope achieves a significantly higher accuracy than the probe on six out of twelve tasks ($p < 1e-5$), and works comparably well in all but one of the remaining six. These results suggest that, in the majority of cases, the source representation without its original context carries enough information about many attributes that a targeted Patchscope can extract. We also study how the accuracy changes across the source layers, and observe that Patchscope consistently outperforms the baseline in early layers, outperforms or works on par with the baseline in mid layers, and almost all cases where it performs worse than the baseline occur in later layers. Our interpretation is that given the language modeling training objective, the representations shift toward next-token prediction in the later layers. Therefore, the attribute of interest would not be as readily accessible via the model’s computation in these layers. This interpretation is also aligned with recent findings that show no decline in using linear relational embedding in predicting ω only when the next token also happens to be ω (Hernandez et al., 2023b). Note that this pattern explains the higher standard deviation of Patchscope accuracy observed in Tab. 2. We discuss this phenomenon in more detail in §C (see Fig. 6).

4.3. Analyzing Entity Resolution in Early Layers

The previous sections focused on analyzing the information encoded in a single hidden state. Here we turn to consider a more global question of how LLMs resolve entity mentions across multiple layers. Concretely, given a subject entity name, such as “the summer Olympics of 1996”, how does the model contextualize the input tokens of the entity and at which layer is it fully resolved?

Answering these questions is hard with existing methods; vocabulary projections focus on the output prediction and fail to show clear patterns in early layers, and probing is restricted to outputs from a fixed number of classes, which may not be expressive enough to describe this process. Alternative approaches have studied this process indirectly via interventions (Meng et al., 2022a), showing that the model constructs a subject representation at the last token of the entity name. However, it is still unclear how this